

Weak-Star Sequential Density Passes to Bochner L^p in the RNP Regime

Discovery packet for arXiv:2604.14068
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Status. Candidate substantial partial result, likely valid, requiring expert review. It closes the RNP subcase that Question 3.13 explicitly says remains open, for every probability space and every $1 < p < \infty$. It also settles purely atomic probability spaces without RNP assumptions. The fully general non-atomic question remains open.

1 Source and exact question

The source is Saurabh Dwivedi, *A study on coreflexive Banach spaces*, arXiv:2604.14068; published online in *Proceedings – Mathematical Sciences*, DOI 10.1007/s12044-026-00868-3 [1]. Question 3.13 on page 13 asks:

Is $L^p(\mu, X)$ weak-star sequentially dense in its bidual whenever X is weak-star sequentially dense in its bidual? The source adds that this remains open even when both X^* and X^{**} have the Radon–Nikodym property.

Here X is *weak-star sequentially dense in X^{**}* if every $x^{**} \in X^{**}$ is the $\sigma(X^{**}, X^*)$ -limit of a sequence from the canonical copy of X .

Fact 3.9. *Let X Banach space. If $x \in X_1$ is a w-PC, then it remains a w-PC in all the higher even-order dual unit balls.*

Proof. To see that $x \in X_1^{**}$ is a w-PC, we apply the above proposition to $K = X_1 \subseteq X_1^{**}$. By using this proposition inductively to $K = X_1^{(n)}$ for each even $n \in \mathbb{N}$, one can show that x remains a w-PC in all the higher even-order dual unit balls. $\square$

In the below, we provide an easy alternative proof of Fact 3.9 for a special class of Banach spaces, namely, quasi-reflexive spaces.

Lemma 3.10. *Let X be a quasi-reflexive space. If $x \in X_1$ is a w-PC, then it remains a w-PC in all the higher even-order dual unit balls.*

Proof. Since X is quasi-reflexive, we have $X^{**} = X \oplus F$ for some finite dimensional subspace F of X^{**} . Let $P : X^{**} \rightarrow X^{**}$ be the corresponding contractive projection such that $\ker(P) = F$ and $\text{range}(P) = X$ and let $x \in X_1$ be a w-PC. We first show that it remains a w-PC in X_1^{**} by showing that any net in X_1^{**} converging to x in the w-topology, converges in the norm. Let $\{x_\alpha^{**}\}_{\alpha \in J} \subseteq X_1^{**}$ be a net such that $x_\alpha^{**} \rightarrow x$ in the w-topology. Since P is weakly continuous, we have that $P(x_\alpha^{**}) \rightarrow x$ in the w-topology, so in the norm. Note that the net $\{x_\alpha^{**} - P(x_\alpha^{**})\}_{\alpha \in J} \subseteq F$ and converges to 0 in the w-topology. Since F is finite dimensional, we have $x_\alpha^{**} - P(x_\alpha^{**}) \rightarrow 0$ in norm. By adding the last two norm-convergences we get $x_\alpha^{**} \rightarrow x$ in the norm. A similar line of reasoning shows that x retains the w-PC property in every higher even-order dual unit ball. $\square$

The following note outlines several open problems that are motivated by our earlier results.

Question 3.11. *Conclusion of Theorem 2.5 is unknown without assuming that X is w^* -sequentially dense in X^{**} .*

Question 3.12. *It is not known whether the conclusion of Theorem 3.3 holds without assuming either that both X^* and X^{**} satisfy the R.N.P., or that X^{**}/X is a separable space.*

Question 3.13. *It is not known whether the space $L^p(\mu, X)$ is w^* -sequentially dense in its bidual whenever X is w^* -sequentially dense in its bidual. Even in the case, where both X^* and X^{**} have the R.N.P., This question remains open.*

Figure 1: The three final questions in arXiv:2604.14068, page 13. Question 3.13 is the target of this packet.

2 Result

Theorem 1 (RNP and atomic subcases of Question 3.13). *Let $1 < p < \infty$, let q be conjugate to p , and suppose that X is weak-star sequentially dense in X^{**} .*

- (a) *If X^* and X^{**} have the Radon–Nikodym property, then for every probability space (Ω, Σ, μ) , the canonical copy of $L^p(\mu, X)$ is weak-star sequentially dense in $L^p(\mu, X)^{**}$.*
- (b) *If (Ω, Σ, μ) is purely atomic, then the same conclusion holds with no Radon–Nikodym assumptions on X^* or X^{**} .*

Part (a) is exactly the subcase singled out as still open in the final sentence of Question 3.13. No coreflexivity assumption is needed. Part (b) is also an instance of the source’s preceding ℓ_p -sum proposition; it is included because Lemma 2 supplies the uniform boundedness needed for a fully dominated coordinate proof. The novelty claim in this packet concerns part (a), not that same-paper atomic consequence.

3 Proof intuition

The definition gives a bounded approximating sequence for each individual x^{**} , but apparently with a bound depending on the point. The first key observation is that the dependence is automatically uniform. Put on X^{**} the infimum of the sup norms of all weak-star convergent lifting sequences. This “sequential approximation norm” is complete. The open mapping theorem therefore makes it equivalent to the ordinary bidual norm.

Under the two RNP assumptions, the standard Bochner duality identifications are

$$L^p(\mu, X)^* = L^q(\mu, X^*), \quad L^p(\mu, X)^{**} = L^p(\mu, X^{**}).$$

Write an arbitrary $f \in L^p(\mu, X^{**})$ as a norm-convergent series of simple increments whose L^p norms are summable. Lift every value of every increment by its uniformly bounded weak-star approximating sequence. At stage n , add the n th lifts of the first n increments. Dominated convergence first on each finite-range increment and then on the summable increment index proves weak-star convergence against every $L^q(\mu, X^*)$ test function.

On a purely atomic probability space, Bochner L^p is a weighted countable ℓ_p -sum. The same uniform constant and an ordinary dominated-series argument work coordinatewise, without any RNP representation theorem.

4 Uniform sequential lifting

Lemma 2 (Sequential approximation norm). *Suppose X is weak-star sequentially dense in X^{**} . There is a constant $K_X < \infty$ such that, for every $x^{**} \in X^{**}$, one can choose a sequence $(x_n) \subset X$ satisfying*

$$x_n \xrightarrow{w^*} x^{**}, \quad \sup_n \|x_n\| \leq K_X \|x^{**}\|. \tag{1}$$

Proof. For $z \in X^{**}$ define

$$|z|_s = \inf \left\{ \sup_n \|z_n\| : (z_n) \subset X, z_n \xrightarrow{w^*} z \right\}. \tag{2}$$

The hypothesis makes this finite everywhere. It is a norm: homogeneity and the triangle inequality follow by taking scalar multiples and termwise sums of lifting sequences, while weak-star lower semicontinuity of the norm gives $\|z\| \leq |z|_s$.

We verify completeness. Let (z^m) be Cauchy in $|\cdot|_s$. It is Cauchy in the ordinary norm, so let z be its norm limit. Pass to a subsequence, still denoted (z^m) , such that

$$|z^{m+1} - z^m|_s < 2^{-m-2}.$$

Put $u_m = z^{m+1} - z^m$. Choose $u_{m,n} \in X$ with $u_{m,n} \rightarrow u_m$ weak-star and $\sup_n \|u_{m,n}\| \leq 2^{-m-1}$. Also choose a bounded weak-star lifting $(a_n) \subset X$ of z^1 . Then

$$v_n = a_n + \sum_{m=1}^n u_{m,n} \in X$$

is bounded. For each $x^* \in X^*$, dominated convergence over the counting measure on m gives

$$x^*(v_n) \longrightarrow z^1(x^*) + \sum_{m=1}^{\infty} u_m(x^*) = z(x^*).$$

Indeed, for fixed N the triangular sums

$$w_{N,n} = \sum_{m=N}^n u_{m,n} \quad (n \geq N)$$

belong to X , have norms at most $\sum_{m=N}^{\infty} 2^{-m-1}$, and converge weak-star to $\sum_{m=N}^{\infty} u_m = z - z^N$ by the same dominated convergence argument. Consequently,

$$|z - z^N|_s \leq \sum_{m=N}^{\infty} 2^{-m-1},$$

so the selected subsequence converges in $|\cdot|_s$; the original Cauchy sequence does as well. Thus $(X^{**}, |\cdot|_s)$ is Banach.

The identity

$$I : (X^{**}, |\cdot|_s) \longrightarrow (X^{**}, \|\cdot\|)$$

is a continuous linear bijection. The open mapping theorem gives $|z|_s \leq C_X \|z\|$. Taking a sequence within $\|z\|$ of the infimum in (2) gives (1) with $K_X = C_X + 1$ (and the zero sequence when $z = 0$). $\square$

5 Proof of Theorem 1

The RNP case

Assume X^* and X^{**} have the RNP. Standard Bochner duality [2, Chapter IV] gives the canonical isometric identifications

$$Y := L^p(\mu, X), \quad Y^* = L^q(\mu, X^*), \quad Y^{**} = L^p(\mu, X^{**}). \quad (3)$$

Fix $f \in L^p(\mu, X^{**})$. Choose simple functions s_m such that $s_0 = 0$ and

$$\|f - s_m\|_p \leq 4^{-m} \quad (m \geq 1).$$

Set $d_m = s_m - s_{m-1}$. Then every d_m is simple,

$$f = \sum_{m=1}^{\infty} d_m \quad \text{in } L^p(\mu, X^{**}), \quad \sum_{m=1}^{\infty} \|d_m\|_p < \infty. \quad (4)$$

Write the finite range of d_m as $\{v_{m,1}, \dots, v_{m,r_m}\}$. By Lemma 2, for every nonzero $v_{m,j}$ choose $x_{m,j,n} \in X$ with

$$x_{m,j,n} \xrightarrow{w^*} v_{m,j}, \quad \sup_n \|x_{m,j,n}\| \leq K_X \|v_{m,j}\|,$$

and use zero for a zero value. Replacing every value $v_{m,j}$ of d_m by $x_{m,j,n}$ gives an X -valued simple function $d_{m,n}$ with

$$d_{m,n}(t) \xrightarrow{w^*} d_m(t), \quad \|d_{m,n}(t)\| \leq K_X \|d_m(t)\| \quad (t \in \Omega). \quad (5)$$

Define

$$g_n = \sum_{m=1}^n d_{m,n} \in L^p(\mu, X). \quad (6)$$

Let $h \in L^q(\mu, X^*)$. For fixed m , pointwise weak-star convergence in (5), followed by dominated convergence and Hölder's inequality, gives

$$\int_{\Omega} \langle d_{m,n}(t) - d_m(t), h(t) \rangle d\mu(t) \longrightarrow 0. \quad (7)$$

The absolute value of the integral in (7) is at most $(K_X + 1)\|d_m\|_p \|h\|_q$. These majorants are summable by (4). Treating the omitted terms $m > n$ as $-\int \langle d_m, h \rangle$ and applying dominated convergence once more, now over the counting measure on m , yields

$$\int_{\Omega} \langle g_n(t) - f(t), h(t) \rangle d\mu(t) \longrightarrow 0.$$

By (3), this says exactly that $g_n \rightarrow f$ in $\sigma(Y^{**}, Y^*)$. Part (a) follows.

The purely atomic case

A purely atomic probability space gives, after rescaling the atoms, an isometric identification

$$L^p(\mu, X) \cong \left(\bigoplus_{k \in I} X \right)_{\ell_p},$$

where I is finite or countable. For $1 < p < \infty$, without any RNP assumption,

$$\left(\bigoplus_{k \in I} X \right)_{\ell_p}^{**} = \left(\bigoplus_{k \in I} X^{**} \right)_{\ell_p}, \quad \left(\bigoplus_{k \in I} X \right)_{\ell_p}^* = \left(\bigoplus_{k \in I} X^* \right)_{\ell_q}.$$

Given $z = (z_k) \in \ell_p(I; X^{**})$, use Lemma 2 to choose $x_{k,n} \rightarrow z_k$ weak-star with $\sup_n \|x_{k,n}\| \leq K_X \|z_k\|$, and put $x^{(n)} = (x_{k,n})_{k \in I}$. Then $x^{(n)} \in \ell_p(I; X)$ and $\|x^{(n)}\|_p \leq K_X \|z\|_p$. Against every $x^* = (x_k^*) \in \ell_q(I; X^*)$, coordinatewise convergence and the summable majorant $2K_X \|z_k\| \|x_k^*\|$ give

$$\sum_{k \in I} \langle x_{k,n}, x_k^* \rangle \longrightarrow \sum_{k \in I} \langle z_k, x_k^* \rangle.$$

This proves part (b). □

6 Upgrade attempts and remaining obstruction

Eight materially distinct routes were checked on this target.

- Route 1.** The sequential approximation gauge was introduced and proved complete; the open mapping theorem supplied the uniform lifting constant. This route succeeded and is the reusable core of the result.
- Route 2.** A direct lift of one simple approximation at a time fails because a single diagonal choice need not control the nonmetrizable weak-star topology. Replacing approximations by a summable series of simple increments fixes the problem and proves the RNP subcase.
- Route 3.** The coordinate version of the same uniform bound proves the purely atomic case without RNP assumptions.
- Route 4.** If X^* has the RNP but X^{**} does not, then $L^p(\mu, X)^* = L^q(\mu, X^*)$, but the bidual can contain nonrepresentable functionals beyond $L^p(\mu, X^{**})$. The proof reaches the canonical representable part only.
- Route 5.** If X^{**} has the RNP but X^* does not, the dual itself contains nonrepresentable vector-measure functionals. Pointwise weak-star convergence in X^{**} does not control those extra tests.
- Route 6.** A measurable-selection approach was tested. It cannot address a general element of $L^p(\mu, X)^{**}$ without the RNP identifications, because such an element has no pointwise X^{**} values from which to select lifts.
- Route 7.** Finite partitions and martingale conditional expectations approximate the representable Bochner subspace, but singular bidual components persist under refinement. No sequential lifting of those singular martingale differences follows from the hypothesis on X .
- Route 8.** Counterexample routes through $c_0(\Gamma)$, ℓ_1 , and complemented copies of ℓ_1 were checked. The obvious nonseparable candidates already fail weak-star sequential density, while that property passes to closed subspaces and excludes ℓ_1 . No viable counterexample survived.

The exact remaining obstacle is therefore the singular part of the Bochner bidual. The hypothesis controls sequences in X^{**} , while without RNP a general point of $L^p(\mu, X)^{**}$ need not be an X^{**} -valued function at all.

7 Verification, novelty bounds, and recommendation

The proof was audited at three levels: completeness of the approximation gauge, the two dominated-convergence passages in the RNP argument, and the unconditional coordinate domination in the atomic argument. No computation is used. The only external mathematical input beyond standard Banach-space theorems is the classical RNP representation of Bochner duals.

The four lightweight run indexes were searched for arXiv:2604.14068, “coreflexive,” “weak-star sequentially dense,” “Bochner L^p ,” and the exact final-question phrasing. A bounded arXiv search returned only the source for “coreflexive Banach,” while Crossref/OpenAlex searches for the weak-star/Bochner combination returned no matching result. OpenAlex listed zero citing works for

the published source on 11 August 2026. The 1970 paper of McWilliams on weak-star sequential density was also checked at the metadata level; no Bochner-space answer was located. This is not an exhaustive literature certification.

Human review should focus first on Lemma 2, especially the counting-measure dominated convergence used to prove completeness, and then on the double domination in (7). If those steps are accepted, part (a) directly closes the precise RNP subcase emphasized by the source and should be recorded as a substantial partial answer.

References

- [1] S. Dwivedi, *A study on coreflexive Banach spaces*, arXiv:2604.14068 (2026); Proceedings – Mathematical Sciences, DOI 10.1007/s12044-026-00868-3.
- [2] J. Diestel and J. J. Uhl, Jr., *Vector Measures*, Mathematical Surveys 15, American Mathematical Society, 1977.